



Internship Report Form

Student-Intern's Name	:	Skyler Jason P. Banzon				
Name of HTE	:	Cubeworks Technology Consulting and Solutions, Inc.				
HTE Supervisor's Name	:	Elmer N. Canque (CIO)				
Report Period	From	:	March 2, 2026	To	:	March 15, 2026
Number of Hours Worked This Quarter	:	80				
Cumulative No. of Hours to Date	:	80				

Quarter Summary

No.	Tasks Performed	% Time Spent
1	Company orientation, workstation and development environment setup, and introduction to industrial automation tools and internal workflows	10%
2	Study and configuration of industrial communication protocols (TCP/IP, Modbus) for enabling data exchange between Node-RED and factory simulation environments	15%
3	Database integration and process data logging: setting up PostgreSQL, designing data schemas, and connecting database nodes within Node-RED for persistent storage of machine data	15%
4	Development and testing of practice machine simulations in Factory I/O: building conveyor-based automation sequences with sensor integration, controlled via Node-RED over Modbus TCP/IP	15%
5	Node-RED Dashboard UI design and development: creating real-time monitoring interfaces with charts, status indicators, and control elements to visualize live machine states and logged data	15%
6	Development and testing of additional practice machine simulations incorporating state machine logic, timed sequences, cycle-based data logging, and dashboard integration	15%
7	Design and implementation of machine visualization widgets using HTML templates in Node-RED Dashboard to reflect live machine states and process animations	10%
8	Integration testing, debugging, and optimization of all developed flows, dashboards, and database connections to ensure reliable end-to-end system operation	5%
TOTAL		100%



Additional Activities (other growth opportunities e.g., conferences, training, seminars etc.)

- Self-directed study of industrial communication protocols (Modbus TCP/IP, TCP/IP server-client architecture) and PostgreSQL database administration relevant to SCADA systems.

List any holidays, personal days, and/or days missed this quarter:

- None. Full attendance maintained throughout the reporting period (March 2 – March 15, 2026).

Quarter Analysis, Reflection, and Plan of Action

What experiences were particularly rewarding during this reporting period?

Built a full-scale SCADA dashboard integrating Node-RED, Factory I/O, and PostgreSQL. It was a great challenge to bridge the gap between Modbus TCP/IP communication and real-time HTML visualization, building on previous logic-based projects like water tank monitoring and traffic light controllers.

What experiences were particularly difficult during this reporting period?

Building the system required navigating several technical challenges, particularly around Node-RED data logging and identifying the most effective SCADA visualization techniques. Beyond the interface, refining the system logic and overall architecture was a key focus to ensure the real-time charts remained accurate and readable without becoming cluttered as the data scaled.

(If Applicable) Other comments, items, or issues that you want your supervisor or adviser to be made aware of? What is your proposed plan of action to resolve this issue?

No major issues to report. The internship has been progressing smoothly and in alignment with the training plan. All simulations and dashboards developed so far are functional and have been tested with live Factory I/O connections.

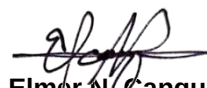


UNIVERSITY *of*
SAN CARLOS
SCIENTIA • VIRTUS • DEVOTIO

Department of Computer Engineering
School of Engineering


Skyler Jason P. Banzon

Student-Intern


Elmer N. Canque

HTE Supervisor

Date Submitted : March 30, 2026

Date Reviewed : March 30, 2026

Supervisor's Comments:

None